EVERY H COVER OF AN AFFINE SCHEME IS REFINED BY A FINITE AFFINE COVER OF A SURJECTIVE PROPER FINITELY PRESENTED MORPHISM

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file flat.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-flat/retained-preamble.tex), a verbatim excerpt of the native Xy-pic setup and of the native theorem declarations and mathematical macros, followed by the project own definitions of the reference, slogan and history display environments (verbatim package plus three newenvironment lines) that the retained file carries at its end and that the project itself typesets as comments. No macro was invented or redefined, and the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

Editorial note. The counted statement and proof contain no cross reference, no citation and no included file, so the delivered body is self contained.

Editorial note. Canonical location of the counted proposition: Stacks tag 0ETU (sources/stacks-flat/pinned-original-source.tex line 10303).

Allowed standard prerequisites (editorial).

- Definition of an h covering (and of the ph covering used in the comparison). Same chapter, Definition labelled definition-h-covering, cited at proof lines 10322-10325, with Topologies, Definition labelled topologies-definition-ph-covering. Supplies the initial proper surjective cover and its refining finite affine cover.
- Approximating a proper morphism by proper finitely presented morphisms. Limits, Lemma labelled limits-lemma-proper-limit-of-proper-finite-presentation, cited at proof lines 10327-10329. Writes Y -> X as a directed limit of proper finitely presented morphisms Y_lambda -> X with surjectivity preserved.
- Descending opens and affineness through a directed limit. Limits, Lemmas labelled limits-lemma-descend-opens and limits-lemma-limit-affine, cited at proof lines 10335-10341. Spreads the finite affine cover to some Y_lambda.
- The functorial characterisation of local finite presentation. Limits, Proposition labelled limits-proposition-characterize-locally-finite-presentation, cited at proof lines 10343-10347. Descends the refinement morphisms h_j to Y_{lambda,j}.

Source reference. Original definition 0ETS: sources/stacks-flat/pinned-original-source.tex, source lines 10171–10171. Same-chapter pointer into the author environment definition at source line 10171; invoked at proof line 10321 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Date: stacks-project:flat.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Source reference. Original definition 0DBG: sources/stacks-chow/tags.txt, source lines 16696–16696. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 10322 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 09ZR: sources/stacks-chow/tags.txt, source lines 12609–12609. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 10326 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 01Z4: sources/stacks-chow/tags.txt, source lines 2579–2579. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 10335 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 01Z6: sources/stacks-chow/tags.txt, source lines 2581–2581. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 10342 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original proposition 01ZC: sources/stacks-chow/tags.txt, source lines 2587–2587. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 10347 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Lemma 0.1. *Let X be an affine scheme. Let $\{X_i \to X\}_{i\in I}$ be an h covering. Then there exists a surjective proper morphism*

$$Y \longrightarrow X$$

of finite presentation (!) and a finite affine open covering $Y = \bigcup_{j=1,\ldots,m} Y_j$ such that $\{Y_j \to X\}_{j=1,\ldots,m}$ refines $\{X_i \to X\}_{i\in I}$.

Proof. By assumption there exists a proper surjective morphism $Y \to X$ and a finite affine open covering $Y = \bigcup_{j=1,\ldots,m} Y_j$ such that $\{Y_j \to X\}_{j=1,\ldots,m}$ refines $\{X_i \to X\}_{i\in I}$. This means that for each j there is an index $i_j \in I$ and a morphism $h_j : Y_j \to X_{i_j}$ over X. See Definition 0ETS and Topologies, Definition 0DBG. The problem is that we don't know that $Y \to X$ is of finite presentation. By Limits, Lemma 09ZR we can write

$$Y = \lim Y_\lambda$$

as a directed limit of schemes Y_λ proper and of finite presentation over X such that the morphisms $Y \to Y_\lambda$ and the the transition morphisms are closed immersions. Observe that each $Y_\lambda \to X$ is surjective. By Limits, Lemma 01Z4 we can find a λ and quasi-compact opens $Y_{\lambda,j} \subset Y_\lambda$, $j = 1, \ldots, m$ covering Y_λ and restricting to Y_j in Y. Then $Y_j = \lim Y_{\lambda,j}$. After increasing λ we may assume $Y_{\lambda,j}$ is affine for all j, see Limits, Lemma 01Z6. Finally, since $X_i \to X$ is locally of finite presentation we can use the functorial characterization of morphisms which are locally of finite presentation (Limits, Proposition 01ZC) to find a λ such that for each j there is a morphism $h_{\lambda,j} : Y_{\lambda,j} \to X_{i_j}$ whose restriction to Y_j is the morphism h_j chosen above. Thus $\{Y_{\lambda,j} \to X\}$ refines $\{X_i \to X\}$ and the proof is complete. □